

THE CHANDIGARH POLICE HACKATHON



TEAM NAME: CyberX

TEAM LEADER NAME: Bhawna Bhadana

INSTITUTION : Delhi Skill and Entrepreneurship University

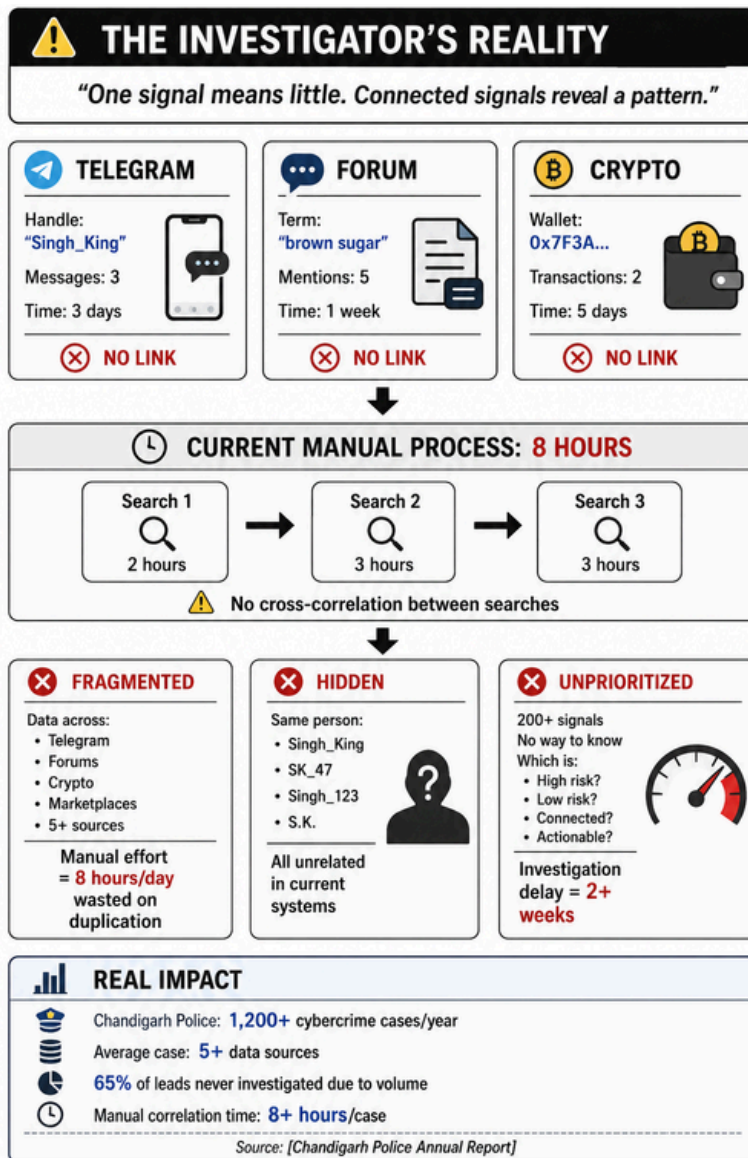
CyberX | Chandigarh Police Hackathon 2026 | Problem Statement 3

TRACE-X : From fragmented signals to verified investigative leads.

An explainable intelligence-fusion platform that connects aliases, entities, relationships and cross-source signals to prioritize evidence-backed investigative leads.

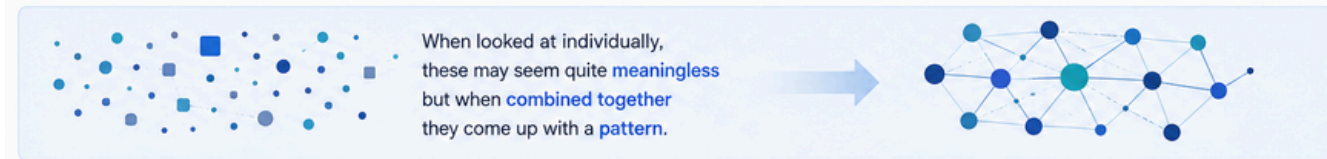
Problem Statement

"One signal may mean little. Connected signals can reveal a pattern."



The main problem is **not data** but the issues surrounding that data.

Investigators need to understand that digital intelligence can come in various forms like **identity, relationships, communication, transactions, topics, date and time**, etc.



PROBLEMS WE FACE TODAY



Challenge

THE INVESTIGATOR'S PROBLEM

1. FRAGMENTED

Intelligence scattered across multiple sources.

2. HIDDEN

Aliases and disconnected identifiers hide relationships.

3. UNPRIORITIZED

Too many signals, no clear lead to investigate first.

Key Features

01 Entity Resolution

Cross-source alias matching, even across investigations, with visible confidence and reasons.

02 Network Intelligence

Interactive relationship graph with live degree-centrality and connection-density metrics.

03 Explainable Risk Scoring

Deterministic, itemized 0–100 priority score with a full "Why this score?" breakdown.

04 Automated Alerts

Rule-based detection from correlated signals → analyst verifies, rejects, or requests more.

05 Evidence Vault

SHA-256 integrity hashing with full Finding → Source → Integrity → Decision lineage.

A SINGLE SYSTEM WITH FIVE LEVELS OF KNOWLEDGE.

1 ENTITY INTELLIGENCE



Detect and connect aliases, IDs, topics, and other entities.

POSSIBLE MATCH



DEGREE OF CERTAINTY



EXPLANATION

- ✓ Name similarity
- ✓ Shared aliases
- ✓ Common source
- ✓ Activity overlap

REVIEW BY THE EXPERT



- Detect aliases, IDs, topics, phones, emails, wallets and more.
- AI suggests possible matches.
- Shows degree of certainty.
- Explains why.
- Expert reviews and decides.

2 DYNAMIC NETWORK INTELLIGENCE



More than a simple relationship chart.



TRACE-X FINDS OUT:

- Who is connected?
- How are they related?
- What clusters are emerging?
- What is the history of the network?

- Real-time relationship mapping.
- Cluster detection.
- Network evolution over time.
- Strong/weak relationship identification.

3 EMERGING THREAT RADAR



Discover change instead of waiting for visible alerts.



IDENTIFIES:

- Spikes in activity
- Emerging clusters
- Fast-growing indicators
- Development of relationships
- Time-related anomalies

- Detects weak signals early.
- Monitors changes over time.
- Highlights unusual patterns.
- Helps act before threats become visible.

4 EXPLAINABLE PRIORITY ENGINE



Instead of saying:
✗ "AI indicates that this is HIGH RISK."

TRACE-X explains the reason:

"Priority level is **82** from 100 and here is the reason for it."



- Every priority score is explainable.
- Investigators see why a case is important.
- Built on multiple intelligence factors.

5 EVIDENCE → INSIGHT → DECISION



Wherever you look at an important discovery it has a transparent path:

SOURCE



Dark Web / Forums / Encrypted Platforms

EVIDENCE



Raw Data / Post / Transaction / Message

ANALYSIS



AI Processing / Correlation / Validation

DISCOVERY



Entity / Relationship / Pattern / Threat

DECISION OF THE EXPERT



Expert Review & Final Decision

VERIFY ✓

REJECT ✗

NEED MORE EVIDENCE ?

06 Trend Radar

Growth and confidence tracking for emerging activity patterns.

07 AI Narration (Groq)

Plain-language explanations of findings — narration only, never a decision.

08 Reporting

Live, database-driven investigation reports, generated and audit-logged on demand.

09 Audit & Access Control

JWT + bcrypt authentication, RBAC, full who / what / when logging.

10 Global Search

Cross-entity, alert and evidence lookup by keyword, alias, wallet address or ID.

Unique Value Proposition

Not a dashboard — an explainable decision-support workflow.

01

EXPLAINABLE BY DESIGN

Every score and match shows its exact reasoning — never a bare black-box percentage.

02

CROSS-INVESTIGATION RESOLUTION

Surfaces the same identity across separate, even archived, cases — most tools stay siloed per case.

03

HUMAN-IN-THE-LOOP, ALWAYS

AI narrates in plain language; deterministic rules score and alert; the analyst verifies and decides.

04

EVIDENCE-TO-DECISION TRACEABILITY

Every finding is hashed, source-linked, and logged end-to-end.

05

BUILT FOR REAL INVESTIGATIVE WORKFLOW

Not just a dashboard — a complete Fragment-to-Report pipeline, live.

EXISTING APPROACH



VS

TRACE-X APPROACH



OUR DIFFERENTIATORS



ENTITY RESOLUTION

Identifies when different records may represent the same real-world entity.



NETWORK-FIRST INVESTIGATION

Moves beyond individual records to understand relationships and networks.



EXPLAINABLE AI

Does not simply provide a score — it shows why an entity or signal is considered important.



EVIDENCE-BACKED INTELLIGENCE

Insights remain traceable to their underlying evidence.



INVESTIGATOR-IN-THE-LOOP

AI assists prioritization and correlation while final verification remains with the investigator.

Explainable AI

Clear and understandable AI insights.

Cross-Investigation

Connects data from multiple sources.

Human-in-the-Loop

Investigator controls final decisions.

Evidence Traceability

Every insight is linked to evidence.

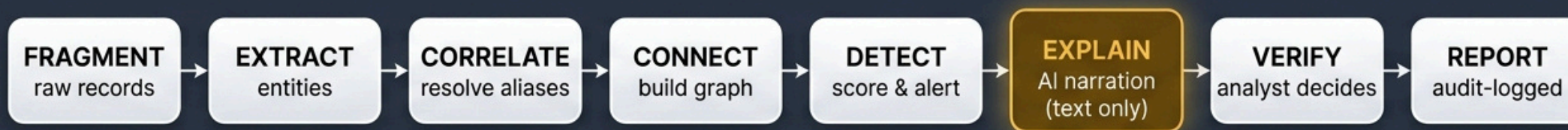
Real Workflow

Built for complete investigation-to-action process.

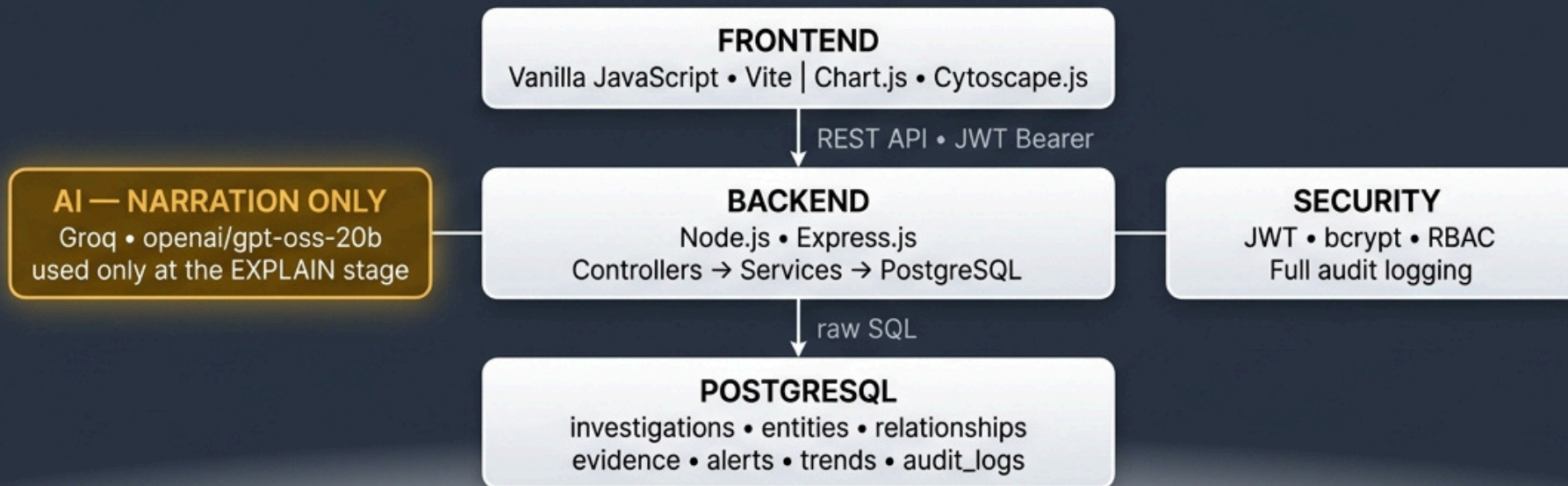
Workflow / Architecture & Tech Stack

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CORE PIPELINE



SYSTEM ARCHITECTURE & TECH STACK



Deterministic SQL and arithmetic everywhere — the AI only narrates the EXPLAIN stage, in plain language, and never decides a score, match, or alert.

LIVE DEMO & NETWORK INTELLIGENCE GRAPH

Autonomous Multi-Source Signal Fusion, AI Copilot Investigation & Force-Directed Link Analysis

TRACE-X Live System Modules

- 01

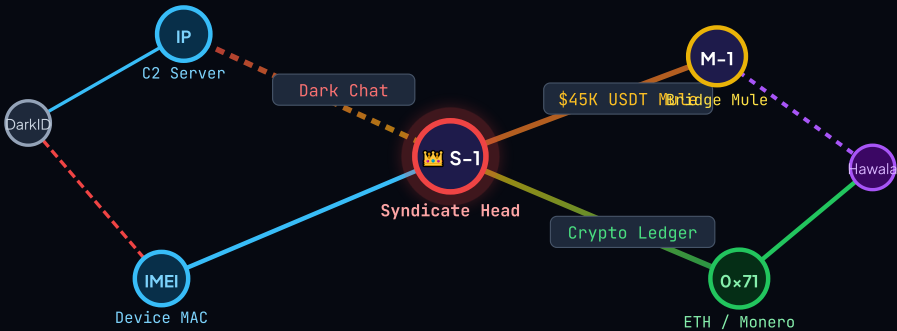
Multi-Source Ingestion & Entity Dossiers
Fuses PCAP metadata, Crypto ledgers, dark-web intercepts & CDR logs into consolidated suspect target dossiers.
- 02

Dual-Engine AI Intelligence Copilot (ꯈꯩꯪ)
Generates instant executive briefings, calculates suspect risk scores, and detects hidden cross-case anomalies.
- 03

Global Geospatial Threat Telemetry
Interactive world map tracking C2 proxy nodes, money laundering routes & data exfiltration in real-time.
- 04

Tamper-Proof Forensic Evidence Vault
Every lead and graph edge is cryptographically secured with SHA-256 hashes for legal court admissibility.

Force-Directed Graph Link Analysis



● Prime Suspect ● Bridge Mule ● Crypto Ledger ● Digital IOC (IP/IMEI)

98.4%
ENTITY ACCURACY

< 120ms
FUSION LATENCY

6+ Hops
TRAVERSAL DEPTH

SHA-256
EVIDENCE HASH

Feasibility and Viability

 FEASIBILITY Can we build it? Yes.		
	Technical Feasibility • Built with proven, reliable technologies. • PostgreSQL for data storage and relations. • Python / FastAPI for backend services. • NetworkX for graph analytics. • LLM (OpenAI) used only for explanation (not decision).	 High
	Operational Feasibility • Aligns with investigator workflows. • Intuitive UI for search, graph, alerts, and reports. • Analyst remains in control at every step.	 High
	Data Feasibility • Works with structured, semi-structured and unstructured digital records. • Supports multiple sources (marketplaces, wallets, forums, communications, etc.). • Data lineage and integrity ensured.	 High
	Legal & Ethical Feasibility • Designed for lawful investigative use. • Maintains audit trails and data integrity. • Protects privacy and follows compliance standards.	 High
	Overall Feasibility Summary All core components are technically achievable, operationally practical, and legally responsible.	HIGHLY FEASIBLE

 VIABILITY Should we build it? Absolutely.		
	Problem Viability • Law enforcement faces real challenges in connecting fragmented digital evidence. • Rising cybercrime, darknet markets, and encrypted communication demand better tools. • Strong need for explainable, auditable intelligence.	 Strong
	Market / User Viability • Primary users: Law enforcement, cybercrime units, intelligence analysts. • Directly supports investigations and decision-making. • High value for agencies handling digital crimes.	 Strong
	Financial Viability • Cost-effective: built on open-source stack. • Scalable deployment (on-premise or cloud). • Long-term savings through faster, accurate investigations. • Potential for licensing to law enforcement agencies.	 Strong
	Strategic Viability • Strengthens national security and public safety. • Promotes data-driven, transparent investigations. • Future-ready platform with AI explainability built-in.	 Strong
	Overall Viability Summary High impact, strong user need, cost-effective, and strategically valuable.	HIGHLY VIABLE

Deliverables / Proposed outcome

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1. DELIVERABLES (What We Deliver)



1. Working Web Platform

- Fully functional intelligence fusion platform
- Secure login & role-based access



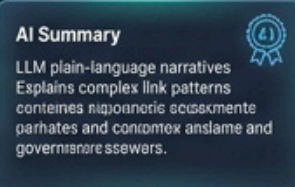
2. Intelligence Graph

- Interactive entity & relationship graph
- Spot hidden connections instantly



3. Risk Scoring Engine

- Transparent multi-factor scoring
- Clear evidence-backed breakdown



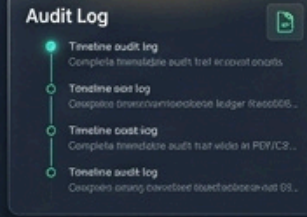
4. AI-Powered Explanation

- LLM plain-language narratives
- Explains complex link patterns



5. Evidence Management

- Cryptographic hash verification
- Admissible & tamper-evident



6. Audit & Case Reports

- Complete immutable audit trail
- PDF/CSV case reports

2. PROPOSED OUTCOME (Impact We Create)



Faster Investigations

Cuts analysis time via automated correlation



Better Decision-Making

Evidence-backed prioritization



Transparent & Defensible

Audit-ready and explainable



Stronger Collaboration

Facilitates seamless cross-agency intelligence sharing



Data Integrity & Trust

Ensures tamper-proof digital evidence



Public Safety Impact

Rapidly dismantling illicit networks

OVERALL IMPACT

TRACE-X empowers investigators with a connected, explainable, and auditable intelligence platform that turns fragmented digital records into actionable, defensible leads — while keeping humans in control.

“ We don't replace investigators. We make them stronger. ”

References



1. RESEARCH PAPERS & PUBLICATIONS

- Al Hasan, M., Chaoji, V., Salem, S., & Zaki, M. (2006). Link Prediction using Supervised Learning. SDM 2006.
- Shen, H., et al. (2019). Explainable AI in Criminal Justice: A Survey. arXiv: 1904.00985.
- Ferrara, E. (2018). The Rise of Complex Social Networks. Nature Physics, 14(11), 1063–1064.
- Lazer, D., et al. (2014). Computational Social Science. Science, 323(5915), 721–723.



2. FRAMEWORKS & TECHNOLOGIES

- Frontend: React.js, Tailwind CSS, Chart.js, React Flow
- Backend: Node.js, Express.js
- Database: PostgreSQL
- Graph Visualization: Cytoscape.js / D3.js
- AI / LLM Integration: OpenAI API (GPT-4o)



3. AI / ML & ALGORITHMS

- Entity Resolution: String Similarity, Blocking, Rule-based Matching
- Link Prediction: Adamic-Adar, Common Neighbors
- Anomaly Detection: Isolation Forest, Z-Score
- Risk Scoring: Weighted Scoring Model (Custom)
- NLP for Explanation: GPT-4o (OpenAI)



4. TOOLS & PLATFORMS

- Postman – API Testing
- Docker – Containerization
- Git & GitHub – Version Control
- VS Code – Development Environment
- Render / Railway – Deployment & Hosting



5. DATA SOURCES & OPEN DATASETS

- EC3 (FBI). (2023). Internet Crime Report.
- UNODC. (2022). World Drug Report.
- Kaggle. (Various). Cybersecurity & Darknet related datasets.
- Google Open Source Threat Intelligence (MISP Feeds).
- Blockchain datasets from public explorers (e.g., Blockchair, Etherscan).



6. LEGAL & POLICY REFERENCES

- Information Technology Act, 2000 (India)
- Digital Personal Data Protection Act, 2023 (India)
- EU GDPR (General Data Protection Regulation)
- NIST SP 800-53: Security and Privacy Controls for Information Systems
- INTERPOL Guidelines for Digital Evidence (2021)



7. CYBERSECURITY STANDARDS

- OWASP Top 10 – 2021
- ISO/IEC 27001: Information Security Management
- NIST Cybersecurity Framework (CSF) 2.0
- CIS Controls v8



8. DOMAIN & SUBJECT MATTER REFERENCES

- Darknet Market Monitoring Reports (Various OSINT Sources)
- Cryptocurrency Analysis Reports (Chainalysis, TRM Labs)
- Law Enforcement Use Cases & Whitepapers
- Academic lectures and notes on Network Science & Graph Theory



We stand on the work of many researchers, practitioners, and open-source communities. TRACE-X is our contribution to building a safer digital world.



NOTE

All references are used strictly for academic and research purposes. TRACE-X does not collect or store real illicit data.